

ECN 164 International Finance — Practice Mock Exam B

Answer Key · Unofficial self-study practice based on Assignment 1, Part I (Chapter 2: quotes, cross rates, bid–ask spreads, triangular arbitrage)

1. (5 points) Flipping a quote. On October 2, 2026 the New Zealand dollar was worth about USD0.5850/NZD and the Swiss franc about USD1.2500/CHF.

- How many New Zealand dollars does one U.S. dollar buy (NZD/USD)?
- How many Swiss francs does one U.S. dollar buy (CHF/USD)?
- Which currency is the “base” currency in each of the original quotes?

Solution

- $\text{NZD/USD} = 1 / 0.5850 = \text{NZD1.7094/USD}$.
- $\text{CHF/USD} = 1 / 1.2500 = \text{CHF0.8000/USD}$.
- The base currency is the one being priced (in the denominator): the **NZD** in USD/NZD and the **CHF** in USD/CHF. The USD is the quote (pricing) currency in both.

2. (5 points) Paying in a foreign currency. You are traveling in Mexico and your hotel costs MXN2,400 per night. The exchange rate is MXN18.50/USD.

- What is the cost of one night in dollars?
- A week later the rate is MXN18.20/USD. Did the peso appreciate or depreciate against the dollar? What does the same room cost in dollars now?

Solution

- $2,400 / 18.50 = \text{\$129.73}$ per night.
- One dollar now buys fewer pesos ($18.50 \rightarrow 18.20$), so the peso **appreciated** against the dollar. New cost: $2,400 / 18.20 = \text{\$131.87}$ — the room is more expensive for a dollar holder.

3. (5 points) Lion City Electronics, Inc. needs to buy 3,000,000 Singapore dollars (SGD) to pay its Singaporean supplier. Its banker quotes bid–ask rates of SGD1.2840–1.2850/USD. What will be the dollar cost of the SGD3,000,000?

Solution

The quote prices the dollar: the bank buys USD at SGD1.2840 (bid) and sells USD at SGD1.2850 (ask). The company buys SGD by *selling* dollars to the bank, so it gets the bank's bid for dollars — the rate that gives the company fewer SGD per dollar.

Dollar cost = $3,000,000 / 1.2840 = \text{\$2,336,448.60}$.

(Check: using 1.2850 would give \$2,334,630.35, a lower cost — the bank never gives the customer the better rate.)

4. (7 points) **Rates of change.** The dollar was CHF0.8050/USD on September 25, 2026 and CHF0.7930/USD one week later.

- By what percentage did the U.S. dollar appreciate or depreciate against the Swiss franc?
- By what percentage did the Swiss franc appreciate or depreciate against the dollar? (Hint: first express the rates as USD/CHF.)
- Why are your answers to (a) and (b) not exactly the same size?

Solution

- $(0.7930 - 0.8050) / 0.8050 = -1.49\%$: the USD **depreciated** by 1.49%.
- USD/CHF: $1/0.8050 = 1.2422$ and $1/0.7930 = 1.2610$. Change = $(1.2610 - 1.2422) / 1.2422 = +1.51\%$: the franc **appreciated** by 1.51%.
- Each percentage change is measured relative to a different base (the starting price of a different currency). Since CHF/USD and USD/CHF are reciprocals, $1 + \%\Delta(\text{USD/CHF}) = 1 / (1 + \%\Delta(\text{CHF/USD}))$, which is not linear, so the two changes are close but not equal in size.

5. (8 points) **Cross rates.** Use the following quotes against the U.S. dollar: USD1.33/GBP, USD0.585/NZD, INR88.20/USD, MXN18.40/USD.

- Calculate the cross rate between the New Zealand dollar and the pound, in NZD/GBP and in GBP/NZD.
- Calculate the rupee–pound cross rate (INR/GBP).
- Calculate the peso–New Zealand dollar cross rate (MXN/NZD).
- With 10 currencies, how many exchange rates would dealers need to quote if every pair traded directly? How many if every currency traded only against the dollar? What does this tell you about the role of the dollar?

Solution

- $\text{NZD/GBP} = (\text{USD/GBP}) / (\text{USD/NZD}) = 1.33 / 0.585 = 2.2735$; $\text{GBP/NZD} = 0.585 / 1.33 = 0.4398$.
- $\text{INR/GBP} = (\text{INR/USD}) \times (\text{USD/GBP}) = 88.20 \times 1.33 = 117.3060$.
- $\text{MXN/NZD} = (\text{MXN/USD}) \times (\text{USD/NZD}) = 18.40 \times 0.585 = 10.7640$.
- All pairs: $N(N-1)/2 = 10 \times 9/2 = 45$ rates. Only against the dollar: $N-1 = 9$ rates. Concentrating trade in one “vehicle currency” (the dollar) makes each of those markets far more liquid and cheap; any cross rate can be built from two dollar quotes.

6. (7 points) Bid–ask spreads in the interbank market. A dealer quotes the Australian dollar at USD0.6548–0.6551/AUD.

- Which rate is the bid and which is the ask? Which rate do you get if you buy Australian dollars from the dealer? If you sell Australian dollars to the dealer?
- What is the spread in pips? What is the spread as a percentage of the ask price?
- A company buys AUD5,000,000 from the dealer and, after changing its mind, sells them back to the dealer a minute later at the same quotes. How many dollars does the company lose?

Solution

- Bid = 0.6548 (the dealer buys AUD), ask = 0.6551 (the dealer sells AUD). You **buy** AUD at the ask, USD0.6551; you **sell** AUD at the bid, USD0.6548.
- Spread = $0.6551 - 0.6548 = 0.0003 = 3 \text{ pips}$. Percentage spread = $0.0003 / 0.6551 = 0.0458\%$.
- Pays $5,000,000 \times 0.6551 = \$3,275,500.00$; receives $5,000,000 \times 0.6548 = \$3,274,000.00$. Loss = **\$1,500.00** (= AUD5,000,000 $\times$ 0.0003).

7. (6 points) Bid–ask spreads at the airport. At the airport, a currency kiosk buys Canadian dollars at USD0.68/CAD and sells Canadian dollars at USD0.77/CAD.

- What is the percentage spread? Compare it with your answer to Problem 6(b).
- You exchange \$800 into Canadian dollars at the kiosk, then change your mind and exchange all of them back into U.S. dollars. How many U.S. dollars do you end up with?
- Give two reasons why spreads for cash (“physical exchange”) are so much larger than interbank spreads.

Solution

- $(0.77 - 0.68) / 0.77 = 11.69\%$, roughly 255 times the interbank spread of 0.0458%.
- Buy CAD at the ask: $800 / 0.77 = \text{CAD}1,038.96$. Sell them at the bid: $1,038.96 \times 0.68 = \text{\$}706.49$ (a loss of \$93.51).
- Any two of: (1) cash transactions are small, so the fixed cost of each trade (staff, rent at the airport, the kiosk's own profit) is spread over few dollars; (2) physical currency must be stored, shipped, insured and protected, which is costly; (3) cash trading is much less liquid/competitive than the interbank market (travelers have few alternatives at the airport), so dealers face more inventory risk and can charge more.

8. (5 points) Flipping a bid–ask quote. A bank quotes the dollar at MXN18.38–18.42/USD. What are the bank's bid and ask prices for the Mexican peso, quoted in USD/MXN? (Hint: remember that the bank always buys low and sells high.)

Solution

When the bank buys MXN it is selling USD, which it does at its USD ask (MXN18.42 per dollar); when it sells MXN it is buying USD at its USD bid (MXN18.38).

Bid for MXN = $1 / 18.42 = \text{USD}0.054289/\text{MXN}$; ask for MXN = $1 / 18.38 = \text{USD}0.054407/\text{MXN}$.

Check: bid < ask, so the bank still buys low and sells high.

9. (6 points) Cross rates with bid–ask spreads. A bank quotes USD0.6548–0.6551/AUD and MXN18.38–18.42/USD. A customer wants to trade Australian dollars for pesos, and the bank will go through the dollar.

a. What is the bank's bid rate for Australian dollars in terms of pesos (MXN/AUD)?

b. What is the bank's ask rate for Australian dollars in terms of pesos (MXN/AUD)?

Solution

a. Bank buys AUD: the customer sells AUD for USD at the AUD bid (0.6548), then sells those USD for MXN at the USD bid (18.38). MXN/AUD bid = $0.6548 \times 18.38 = \mathbf{12.035224}$.

b. Bank sells AUD: the customer buys USD with MXN at the USD ask (18.42), then buys AUD at the AUD ask (0.6551). MXN/AUD ask = $0.6551 \times 18.42 = \mathbf{12.066942}$.

10. (6 points) A Mexican avocado exporter receives AUD2,000,000 from an Australian customer and converts it into pesos with the bank, using the quotes in Problem 9.

a. Which MXN/AUD rate applies, and how many pesos does the exporter receive?

b. How many fewer pesos does the exporter receive than it would at the mid-rates of both quotes?

Solution

a. The exporter sells AUD to the bank, so it gets the bank's **bid** of MXN12.035224/AUD. Proceeds = $2,000,000 \times 12.035224 = \mathbf{MXN24,070,448.00}$.

b. Mid-rates: USD0.65495/AUD and MXN18.40/USD give MXN12.051080/AUD, so the proceeds would be MXN24,102,160.00. Loss from the two spreads = **MXN31,712.00**.

11. (7 points) Cross rates over time. On September 1, 2026 the quotes were USD1.3400/GBP and INR87.50/USD. On October 1, 2026 they were USD1.3150/GBP and INR88.40/USD.

a. Calculate the INR/GBP cross rate on both dates.

b. By what percentage did the pound appreciate or depreciate against the rupee?

c. By what percentage did the rupee appreciate or depreciate against the pound?

Solution

a. Sept 1: $1.3400 \times 87.50 = \mathbf{INR117.2500/GBP}$. Oct 1: $1.3150 \times 88.40 = \mathbf{INR116.2460/GBP}$.

b. $(116.2460 - 117.2500) / 117.2500 = \mathbf{-0.86\%}$: the pound **depreciated** by 0.86% against the rupee.

c. GBP/INR: $1/117.2500 = 0.008529$ and $1/116.2460 = 0.008602$. Change = **+0.86%**: the rupee **appreciated** against the pound. (Note that the rupee weakened against the dollar but strengthened against the pound, because the pound fell even more against the dollar.)

12. (7 points) As a foreign exchange trader, you see the following quotes for New Zealand dollars (NZD), U.S. dollars (USD), and Indian rupees (INR):

USD0.5850/NZD INR50.10/NZD INR88.20/USD

Is there an arbitrage opportunity, and if so, how would you exploit it? Compute your profit per \$1,000,000.

Solution

Implied INR/NZD = (INR/USD) $\times$ (USD/NZD) = $88.20 \times 0.5850 = 51.5970$. The quoted rate is INR50.10/NZD < 51.5970, so **yes**: the NZD is too cheap in the rupee market.

Exploit it by buying NZD where it is cheap (with rupees) and selling it where it is expensive (for dollars):

1) USD $\rightarrow$ INR: $1,000,000 \times 88.20 = \text{INR}88,200,000.00$

2) INR $\rightarrow$ NZD: $/ 50.10 = \text{NZD}1,760,479.04$

3) NZD $\rightarrow$ USD: $\times 0.5850 = \$1,029,880.24$

Profit = **\$29,880.24** (2.99%).

13. (10 points) **Triangular arbitrage.** You observe the following quotes (ignore bid–ask spreads): USD1.3300/GBP in New York, USD0.5850/NZD in Wellington, and NZD2.2200/GBP in London.

- What NZD/GBP cross rate is implied by the two dollar quotes?
- Is there an arbitrage opportunity? In which market is the pound “too cheap”?
- You start with \$1,000,000. Describe the three trades you would make and calculate your profit in dollars.
- What will happen to the three exchange rates as many traders do the same thing?

Solution

a. NZD/GBP = (USD/GBP) / (USD/NZD) = $1.3300 / 0.5850 = \mathbf{2.2735}$.

b. Yes. London quotes NZD2.2200/GBP < 2.2735, so the pound is **too cheap in London** (it costs fewer NZD there than it should).

c. 1) Wellington: buy NZD with USD: $1,000,000 / 0.5850 = \text{NZD}1,709,401.71$. 2) London: buy GBP with NZD: $1,709,401.71 / 2.2200 = \text{GBP}770,000.77$. 3) New York: sell GBP for USD: $770,000.77 \times 1.3300 = \$1,024,101.02$. Profit = **\$24,101.02**.

d. Buying NZD in Wellington pushes USD/NZD **up**; buying GBP with NZD in London pushes NZD/GBP **up**; selling GBP in New York pushes USD/GBP **down**. The implied cross rate (USD/GBP)/(USD/NZD) falls while the London rate rises, until they are equal and the profit disappears.

14. (10 points) Triangular arbitrage with bid–ask spreads. Now the quotes are: USD1.3298–1.3302/GBP, USD0.5848–0.5852/NZD, and NZD2.2640–2.2660/GBP. Starting with \$1,000,000, check both possible routes (dollars → New Zealand dollars → pounds → dollars, and dollars → pounds → New Zealand dollars → dollars). Is there an arbitrage profit once you pay the bid–ask spreads?

Solution

Route 1: USD → NZD → GBP → USD

Buy NZD at the ask: $1,000,000 / 0.5852 = \text{NZD}1,708,817.50$. Buy GBP with NZD at the NZD/GBP ask: $/ 2.2660 = \text{GBP}754,111.87$. Sell GBP at the USD/GBP bid: $\times 1.3298 = \mathbf{\$1,002,817.97}$ (profit of \$2,817.97).

Route 2: USD → GBP → NZD → USD

Buy GBP at the ask: $1,000,000 / 1.3302 = \text{GBP}751,766.65$. Sell GBP for NZD at the NZD/GBP bid: $\times 2.2640 = \text{NZD}1,701,999.70$. Sell NZD at the USD/NZD bid: $\times 0.5848 = \mathbf{\$995,329.42}$ (loss of \$4,670.58).

Yes — Route 1 earns \$2,817.97 even after paying all three spreads. The implied NZD/GBP bid–ask is $2.2724\text{--}2.2746$ ($= 1.3298/0.5852$ to $1.3302/0.5848$); the quoted ask 2.2660 is *below* the implied bid 2.2724, so you can buy pounds with NZD directly and sell them through the dollar for more.

15. (6 points) The foreign exchange market. Answer briefly.

- What is the interbank market, and who are its main participants?
- A customer wants to convert New Zealand dollars into Indian rupees. Why does the bank usually route the trade through the U.S. dollar even though the customer then pays two spreads?
- What is “liquidity,” and how is it related to the size of bid–ask spreads?

Solution

- The interbank (wholesale) market is where large commercial and investment banks and dealers trade currencies with each other in very large amounts (often millions of dollars) electronically or by phone, at very narrow spreads. Smaller banks, firms and individuals reach it through these dealers.
- The NZD/INR market is thin, so a direct quote (if one exists) would have a very wide spread. NZD/USD and USD/INR are each far more liquid, so the two narrow spreads combined are usually cheaper than one wide direct spread.
- Liquidity is the ability to buy or sell a large amount quickly without moving the price. The more liquid a market, the easier it is for dealers to offset positions and the more dealers compete, so the smaller the bid–ask spread.